

Multiple Choice

1. **Answer:** B

Options: A) Refraction; B) de-excitation; C) Chemiluminescence; D) Thermal radiation; E) Phosphorescence

Note: Correct option: B - de-excitation

2. **Answer:** A

Options: A) 2.5 amp; B) 0.5 amp; C) 2.8 amp; D) 3.0 amp; E) 1.0 amp

Note: Correct option: A - 2.5 amp

3. **Answer:** E

Options: A) 700 nm; B) 450 nm; C) 750 nm; D) 300 nm; E) 357 nm

Note: Correct option: E - 357 nm

4. **Answer:** A

Options: A) 3.2×10^{10} cm/sec; B) 3.0×10^{10} cm/sec; C) 2.8×10^{10} cm/sec; D) 2.4×10^{10} cm/sec; E) 1.8×10^{10} cm/sec

Note: Correct option: A - 3.2×10^{10} cm/sec

5. **Answer:** A

Options: A) salinity of the water.; B) proximity to the equator.; C) presence of fish in the pond.; D) type of pond vegetation.; E) acceleration due to gravity.

Note: Correct option: A - salinity of the water.

True / False

6. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

7. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

8. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

9. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: ' 0 m/s^2 '.

10. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** 16.67 m. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** high temperatures. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key